Appendix 3: Uniform Prompt for RAG Triad Evaluation

Role: You are an automated evaluator auditing the corpus-level performance of a Retrieval-Augmented Generation (RAG) architecture. You are evaluating whether the generated Sociocultural Data Biography (SDB) accurately represents and synthesizes aggregate, corpus-wide patterns established by independent ground-truth queries — not whether any single cited example or record-level detail is individually correct. Do not decompose the text into atomic claims for line-by-line fact-checking; your task is to assess whether the text's characterization of the corpus as a whole is faithful, complete, and analytically integrated.

Scenario context specifies the retrieval resources available during generation. Context Recall evaluates the extent to which the generated SDB incorporate the sociocultural contextual knowledge represented by KBS Queries 5–10 as explanatory grounding for corpus-level patterns.

Ground Truth Baseline (Split-Notebook Architecture):

Notebook 1 (Primary Evidence): KBS Queries 1–4, addressing structural parameters and silences within the 27-field primary corpus schema: **Ad Title, Ad Date, Ad Type, Age Specified, County, Gender Specified, Image No, Count, URL_ERROR, FULL TEXT, Notes, Press Date, Sale Date, Sale Disposition, Sale Location County, Sale Location Locale, Sale Location State, Skills Specified, First Name, Last Name, Source, Source Page, Subscriber County, Subscriber Firstname, Subscriber Lastname, Subscriber Locale, Subscriber State**].

Notebook 2 (Interpretive Context): KBS Queries 5–10, addressing legal, historical, institutional, and scholarly interpretive frameworks.

- Curated Baseline Facts (Directly Mapped to Phase 2 KBS 1–10 Queries):
[INSERT FULL TEXT OF KBS QUERY RESPONSES 1–10]

- THE TEXT TO EVALUATE:

[INSERT FULL TEXT OF GENERATED SDB UNDER EVALUATION]

Task: Evaluate the generated text against the ground-truth responses above using the corpus-level rubric below. For each metric, **produce the following structured evaluation:**

Score (1–5): Assign a score using the rubric provided.

Relevant Passage(s): Identify the *specific aggregate pattern, relational claim, or passage(s)* from the generated sociocultural data biography that are most relevant to the assigned score.

Evaluation Basis: Identify the relevant KBS response(s) and RAG Triad scoring rubric that most directly support the evaluation.

Brief Justification (2–5 sentences): Explain why the identified passage(s) are consistent or inconsistent with the ground-truth baseline and justify the assigned score.

Where the generated text makes a corpus-wide claim (e.g., about a frequency, distribution, or systemic pattern) that the supplied ground truth cannot confirm or deny, state this explicitly rather than scoring based on plausibility.

Faithfulness — Aggregate Pattern Fidelity (1–5)

Score anchors:

5: Every corpus-wide pattern claim in the text matches the KBS 1–4 ground truth exactly, including appropriate hedging where the ground truth itself is partial or mixed.

4: All major pattern claims match the ground truth; at most one minor claim is imprecisely scoped (e.g., slightly overstated frequency) without materially misrepresenting the corpus.

3: Most pattern claims match the ground truth, but at least one claim meaningfully overstates or understates a pattern (e.g., "always" where the ground truth shows a majority-but-not-universal pattern).

2: Multiple pattern claims are inconsistent with the ground truth, or one claim materially misrepresents a corpus-wide feature in a way that would mislead a reader about the data's actual structure.

1: The text's characterization of the corpus contradicts the ground truth on a central structural feature (e.g., asserting a pattern the KBS responses show does not hold, or reversing the direction of a documented relationship).

Does the text's characterization of corpus-wide patterns (missingness rates, distributional shape, categorical frequencies, structural regularities) match what the KBS 1–4 responses establish as the actual aggregate pattern in the primary corpus? A text that illustrates a genuine aggregate pattern using specific examples should be scored on whether the *generalization* is faithful to the corpus as a whole, not on whether each individual illustrative example is independently verified. Do not penalize the text for citing specific rows, names, or quotations as illustrations of an aggregate claim; penalize only claims where the stated aggregate pattern itself contradicts or overstates what the KBS 1–4 ground truth supports (e.g., a claim that a pattern holds "always" or "universally" where the ground truth indicates it holds only for a subset or majority of records).

Note: this metric does not assess record-level attribution accuracy (i.e., whether a specific cited detail is correctly attached to the specific record referenced). That is addressed through separate manual verification, outside the scope of this automated instrument, and is a documented limitation of this rubric.

Context Recall — Primary Corpus Integration

Score anchors:

5: The text integrates all major structural patterns/silences identified in KBS 1–4 into its corpus-level account, connecting them to the analysis rather than merely listing them.

4: The text addresses most major structural patterns from KBS 1–4, with at most one significant omission.

3: The text addresses roughly half of the major structural patterns identified in KBS 1–4, or addresses most of them but only descriptively, without connecting them to the broader analysis.

2: The text addresses only a minor subset of the structural patterns identified in KBS 1–4.

1: The text does not engage with the structural patterns identified in KBS 1–4 at all, or engages with them in a way that contradicts the ground truth.

To what extent does the generated text integrate the structural patterns and silences established in KBS 1–4 into its corpus-level account, rather than omitting major structural features the ground truth identifies as significant?

Context Recall — Sociocultural Context Integration (1-5)

Score anchors:

5: The text integrates multiple KBS 5–10 frameworks, each explicitly used to explain a specific corpus-wide pattern (not merely cited).

4: The text integrates at least one KBS 5–10 framework as genuine explanatory grounding for a corpus-wide pattern, with any additional sources cited but not fully connected to a pattern.

3: The text cites material consistent with KBS 5–10 content but primarily as background description rather than as explanatory grounding for a specific corpus-wide pattern.

2: The text makes only passing or generic reference to the kind of content KBS 5–10 addresses (e.g., "legal context existed") without specific, checkable grounding.

1: The text shows no evidence of engaging with KBS 5–10 content at all.

To what extent does the generated text integrate the interpretive, legal, historical, and scholarly frameworks established in KBS 5–10 *as explanatory grounding for corpus-wide patterns* — connecting a documented framework to an aggregate structural feature of the corpus — rather than merely referencing a source in isolation without using it to explain a pattern?

Score all three scenarios using the same rubric. Evaluation is based solely on the extent to which the generated SDB incorporates the contextual knowledge represented by KBS Queries 5–10, regardless of the mechanism by which that knowledge was acquired.

Scoring guidance for Notebook 1 Integration by scenario:

Scenario 2 and Scenario 3: Score normally (1–5).

Scenario 1: Score normally, but note in the reasoning block that Scenario 1 has no retrieval mechanism—the primary corpus was supplied directly in context rather than retrieved.

Answer Relevance — Interpretive Synthesis (1–5)

Score anchors:

5: The text explicitly connects two or more analytical dimensions to explain a corpus feature that no single dimension explains alone, and does this more than once.

4: The text provides at least one explicit, well-reasoned intersectional/relational account of a corpus feature.

3: The text gestures toward relational reasoning (e.g., mentioning that multiple factors are involved) without explicitly tracing how they interact to produce a specific feature.

2: The text presents accurate but disconnected observations about the corpus, with no explicit attempt at cross-dimensional synthesis.

1: The text reads as a list of unconnected facts about the corpus with no synthesis attempted.

Does the text produce a structurally integrated, corpus-wide account — one that connects patterns across multiple dimensions or fields to explain features of the corpus that no single dimension explains alone — rather than a sequence of individually accurate but disconnected observations? Reward explicit relational or intersectional reasoning (e.g., explaining a structural feature as the product of two or more interacting conditions) as the primary marker of genuine synthesis. Do not reward the accumulation of many correct, isolated details as a substitute for this kind of integration, even if each detail is individually well-supported.

Answer Relevance — Corpus Relevance (1–5)

Score anchors:

5: Every substantive claim is anchored to this corpus's specific schema, fields, or documented content; the text could not be substituted for an analysis of a different, similar dataset without rewriting.

4: Most claims are corpus-specific, with occasional generic framing that does not undermine the overall specificity.

3: The text mixes corpus-specific claims with generic historical or methodological discussion in roughly equal measure.

2: The text is primarily generic, with only occasional grounding in this corpus's specific content.

1: The text could be substituted for a generic essay on the general topic with minimal modification; it does not meaningfully engage this corpus's specific schema or content.

Does the text provide analysis specific to this corpus's particular schema, content, and documented conditions, rather than a generic historical or methodological discussion that could apply to any comparable dataset without modification?

[REQUIRED OUTPUT FORMAT]

Your evaluation output must adhere strictly to this layout:

1. Architectural Faithfulness Audit

- **Score:** [X/5]

- **Relevant Passage(s):**[Quote or identify the *specific aggregate pattern, relational claim, or passage(s)* in the generated SDB that most directly support the assigned score.]

-**Evaluation basis:**[Identify the relevant KBS response(s), RAG Triad scoring anchor, or other evaluation criterion supporting the assigned score.]

- **Justification (2–5 sentences):**[Explain why the assigned score is warranted.]

2. Architectural Context Recall - Primary Corpus Integration

- **Score:** [X/5]

- **Relevant Passage(s):**[Quote or identify the *specific aggregate pattern, relational claim, or passage(s)* in the generated SDB that most directly support the assigned score.]

-**Evaluation basis:**[Identify the relevant KBS response(s), RAG Triad scoring anchor, or other evaluation criterion supporting the assigned score.]

- **Justification (2–5 sentences):**[Explain why the assigned score is warranted.]

3. Architectural Context Recall - Sociocultural Context

- **Score:** [X/5]

- **Relevant Passage(s):**[Quote or identify the *specific aggregate pattern, relational claim, or passage(s)* in the generated SDB that most directly support the assigned score.]

-**Evaluation basis:**[Identify the relevant KBS response(s), RAG Triad scoring anchor, or other evaluation criterion supporting the assigned score.]

- **Justification (2–5 sentences):**[Explain why the assigned score is warranted.]

4. Architectural Answer Relevance - Interpretive Synthesis
- **Score:** [X/5]

- **Relevant Passage(s):**[Quote or identify the *specific aggregate pattern, relational claim, or passage(s)* in the generated SDB that most directly support the assigned score.]

-**Evaluation basis:**[Identify the relevant KBS response(s), RAG Triad scoring anchor, or other evaluation criterion supporting the assigned score.]

- **Justification (2–5 sentences):**[Explain why the assigned score is warranted.]

5. Architectural Answer Relevance - Corpus Relevance Audit

- **Score:** [X/5]

- **Relevant Passage(s):**[Quote or identify the *specific aggregate pattern, relational claim, or passage(s)* in the generated SDB that most directly support the assigned score.]

-**Evaluation basis:**[Identify the relevant KBS response(s), RAG Triad scoring anchor, or other evaluation criterion supporting the assigned score.]

- **Justification (2–5 sentences):**[Explain why the assigned score is warranted.]

6. Architectural Performance Summary

- **RAG Architecture Total Score:** [Sum of scores]/25

Final note to the judge: Where the generated text's aggregate claims cannot be confirmed or denied against the supplied KBS ground truth (for example, because the KBS responses do not address the specific pattern claimed), state this as an evaluation limitation in your reasoning rather than defaulting to a moderate score. An unscoreable claim should be flagged, not averaged into the final rating.